Justin J. De La Cruz Romero

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SUMMARY

Software developer and founder with 4+ years of professional software development experience, including hardware-sensing projects in 2021 and 2024 - present. Builds and ships complete products across AI, full-stack, and embedded systems, from concept to live deployment.

EXPERIENCE

Nytsu San Juan, PR
Founder September 2025 - Present

- Designing and building JustIn, a wireless scoring and performance-tracking system for competitive fencing. Own the custom PCB design, firmware, BLE communication between units, and the app.
- Ran a two-hour live field test with fencing instructors and club members in August 2026, using JustIn's scoring system for an entire class. All bouts ran successfully, surfacing concrete improvements for the next iteration.
- Applied domain expertise as a former member of Puerto Rico's national epee team and an active competitive fencer on the island to define the problem and prioritize development.

Special Olympics Puerto Rico San Juan, PR
Software Developer (Independent Contractor), Athlete Portal Puerto Rico October 2026 - Present

- Designing and building a web platform to replace paper forms and spreadsheet-based athlete management, with self-managed accounts for athletes, guardians, and coaches.
- Defining requirements and workflows from the organization's registration and renewal processes.
- Designing for accessibility from the start, as the platform serves athletes with intellectual disabilities.
- Working directly with staff to understand how documents, renewals, and registration run day to day before scoping the platform.

INprende San Juan, PR
Software Developer March 2023 - February 2026

- Built and shipped an internal AI assistant using Gemini and Mastra, initially developing the frontend and later contributing to agent logic, backend, prompt/instruction design, and structured proposal generation.
- Delivered full-stack platforms for government and internal clients, including a ticketing and checkout system (responsive frontend, backend services, database synchronization) and public sector platforms for the Puerto Rico Department of Family and Instituto de Cultura de Puerto Rico.
- Received the Founder Award in 2024 for technical ownership, architecture judgment and independent execution as an in-house software developer.
- Received the 2025 Excellence Award for sustained initiative, rapid execution, and consistently contributing beyond assigned responsibilities.

The Walt Disney Company Orlando, FL
Disney College Program June 2022 - November 2022

- Delivered high-quality guest experience in a fast-paced environment across food and beverage operations at Disney's Hollywood Studios.
- Collaborated across teams to resolve issues and maintain operational efficiency.

SMX Services & Consulting Miami, FL
Web Development Consultant August 2021 - February 2022

- Built full-stack features for an insurance company's property insurance underwriting platform using OutSystems, including UI, backend logic, and database work.
- Developed SQL objects and procedures for database stability and data management, and improved application performance through targeted optimization and client-requested feature design.

FirstBank Puerto Rico San Juan, PR
Software Development Intern February 2021 - August 2021

- Collaborated building an application to reconcile Santander accounts data with FirstBank's records following FirstBank's acquisition of Banco Santander Puerto Rico.
- Supported development across multiple departments, including a customer segmentation dashboard for the credit card division, as well as tools for the auto and human resources divisions.

RESEARCH & PROJECTS

SprayCube: Plant Nutrient-Sensing and Dosing Device March - May 2021
Puerto Rico Science, Technology and Research Trust

- Selected as one of 60 participants in a competitive 12-week program granting access to patented NASA technologies including FlashPose, a spacecraft pose-estimation algorithm.
- Built SprayCube, a plant nutrient-sensing and dosing device, using Python, TensorFlow, and an embedded camera system. Achieved strong detection accuracy on potted plants (~80-90%), while accuracy on open-field plantain crops remained lower (~30%) due to limited testing time and pandemic-related access restrictions.

Undergraduate Final Project: Neural Network Traffic Filtering on Android October 2020 - May 2021
Polytechnic University of Puerto Rico

- As part of this project, we used NetGuard, an open-source Android firewall, as the foundation, and replaced its packet-filtering logic with a custom-trained neural network to determine which packet to drop or allow.

Zutto: Gamified Local-Discovery Platform for Small Businesses 2025
Fase1, Puerto Rico Science, Technology & Research Trust

- Completed both stages of the program (Fase1 Lab and Fase1 Intensive), a 12-week incubator under Puerto Rico's Small Business Incubators and Accelerators Program, developing Zutto into a minimum viable product.
- Selected among 150 submissions to present Zutto, a gamified local-discovery and visibility platform for small businesses, to a public audience.

SKILLS

Hardware & Embedded: PCB Design, Firmware Development, Embedded Systems, BLE/Bluetooth Communication, Peer-to-Peer Device Communication, 3D Printing

Software Development: React, React Native, Node.js, JavaScript, TypeScript, Java, Python, SQL, AWS

AI & Machine Learning: LLM APIs (Gemini, Claude), AI Agents (Mastra), Prompt Design, TensorFlow

AI-Assisted Development: Claude Code, Cursor, MCP (Model Context Protocol)

EDUCATION

Polytechnic University of Puerto Rico, San Juan, PR 2021
B.S. Computer Science